

Seeing the City Through AI Glasses: A Spatial Analysis of Recognition, Recording and Regulation in Convergent Urban Media Systems

Introduction

As digital technologies continue to integrate and become increasingly mobile and embedded, the structure of urban life is also quietly changing. Cities are no longer just places where people move and where space is laid out. Cities now rely more on digital infrastructure to transmit information, sense the environment, and process data (McQuire, 2016; Kitchin & Dodge, 2011). Among them, artificial intelligence (AI) systems have become one of the most representative technologies. These systems are widely used in many fields such as medicine, education, transportation, and public services. In the past, they relied on fixed screens and data centers, but now they are gradually entering people's daily lives and appearing in wearable and interactive ways (Couldry & Hepp, 2017; Plantin et al., 2018).

This research chooses AI smart glasses as the object of analysis. This device is not only wearable and connected to the network, but also can interact with users in real time. It represents a way in which converged media can redefine urban space. We use the official media program "Winning with AI+" as a basis, especially an episode about "smart terminals"¹, to explore the new role of smart glasses in public life. How it allows people to be identified by the system, recorded by data, and regulated by algorithms is the focus of this study. These devices not only enhance human visual abilities, but also change how people perceive space, what they pay attention to, and how they interact with the platform (Andrejevic, 2020; Zuboff, 2019).

¹ "Smart Terminals" is an 89-minute episode from the Chinese state media series Winning with AI+, aired on CCTV-2 on May 16, 2025. It focuses on wearable AI technologies such as smart glasses with functions like real-time translation, sign language recognition, and data archiving. As the only episode that centers on bodily interfaces and spatial infrastructures, it serves as a representative case of AI-mediated urban experience. Full episode available at: <https://tv.cctv.com/2025/05/17/VIDEQFSY1UbK7A4uhcFpf02y250517.shtml>

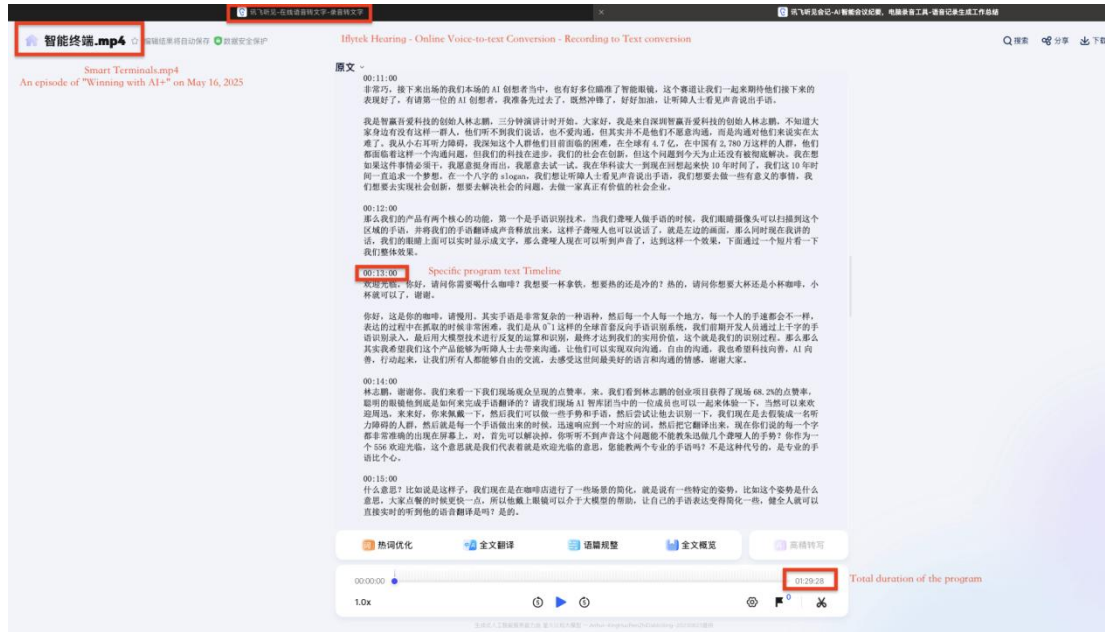


Figure 1. Screenshot of the Video-to-Text Transcript from the Smart Terminals, Created Using iFlytek

In order to more accurately analyze the content of the program, we used iFlytek Hear², a Chinese speech-to-text platform, to transcribe the entire episode minute by minute. The program is 89 minutes long, and we manually proofread the transcription to ensure the accuracy of the time stamp and the reliability of the content. The analysis method adopts discourse analysis (Livingstone, 2004; Van Dijck, 2013), focusing not only on language but also on the expression of images in the program. We believe that this type of media text is not a neutral narrative, but a performative social practice that participates in the construction of the role of artificial intelligence in public life.

To better illustrate how AI glasses work, we propose a three-step framework called “ Recognition-Recording-Regulation ” (R-R-R). This model describes how AI identifies users, records behaviors, and ultimately influences behaviors through algorithmic feedback. These three links are linked together to show how AI devices participate in urban governance. We want to emphasize that media systems not only control the flow of information, but also quietly influence the flow of people, access to space, and the allocation of social resources (Couldry & Mejias, 2019; Mattern, 2017). From this perspective, we can see more clearly that AI glasses are not just a technological product, but also a force that promotes the reorganization of urban social space.

Theoretical Framework

² iFlytek is a Chinese speech recognition platform used in this study for transcribing audiovisual content from national media programs. For more information, visit <https://www.iflytek.com/en/>

The core idea of the research framework is the concept of "media infrastructure". It no longer simply regards media as an information interface but understands it as the underlying system that supports the operation of the city. This system is responsible for coordinating resources, transmitting information, and distributing power (Plantin et al., 2018). Under this understanding, smart glasses are a typical fusion device. It combines perception, data processing, and platform control mechanisms into a wearable medium. This type of technology not only displays the surface of information but also participates in the construction of urban space. As McQuire (2016) said, digital media is now deeply embedded in the daily structure of the city. The convergence here is not just a combination of several technologies but also refers to the close integration of various functions, such as identifying users, calculating data, and regulating behavior, which are now completed by a small device (Jenkins, 2006).

In order to study how this fusion unfolds in urban life, this paper adopts the "recognition-recording-regulation" (R-R-R) framework as an analytical tool. The first stage is "recognition". In a data-centric society, whether the system can "recognize" a person directly determines whether the person can use services, enter spaces, or obtain resources (Andrejevic, 2020; Amoores, 2021). In the scenario of smart glasses, this recognition is mainly achieved through face recognition, voice analysis, or gesture interpretation. The system uses these methods to determine who the user is and what the user should get. This process continues Foucault's (1977) idea that "visibility is power." But unlike the traditional meaning, visibility now often means that the algorithm sets the rules first, and then the system selectively "sees" certain people or behaviors (Cheney-Lippold, 2017). In other words, the system not only recognizes the person himself but also recognizes the "stand-in" composed of data - that is, digital archives, which tell the system how to respond to or restrict the user's actions.

The second stage is "recording." The idea behind this link is that data is not only used to preserve reality, but it itself will also change the appearance of reality. Smart glasses will continuously turn users' vision, hearing, and interactive behaviors into computable data and store these data in the cloud. This information will be classified, organized, and used for future analysis (Van Dijck et al., 2018). This means that technology is no longer just recording the past, but actively "making" digital memories that can be cited and called. In this process, urban space has also become traceable and analyzable. Everyday behavior is like leaving footprints, captured and encoded by the platform, and can be called again at any time (Galič et al., 2017). The "code/space" theory proposed by Kitchin and Dodge (2011) has a good explanation for this: space is no longer just a background, it is "generated" by software and data.

The last stage is "regulation". In this link, the information previously identified and recorded will be used to make decisions. These decisions may be to remind users of certain behaviors, issue alarms, or even directly intervene in the direction of users' actions (Zuboff, 2019; Williamson, 2022). Smart glasses are not passive tools, they can actively influence people's behavior paths. By embedding control means into

devices and interfaces, platforms do not need to "govern" users in the form of external commands, but "silently guide" through recommendations, feedback, and automatic responses. Couldry and Mejias (2019) pointed out that today's platform is like a new governance infrastructure. It not only arranges the market but also affects people's identities, roles, and daily behaviors. Under this architecture, regulation is no longer an institutionalized control from top to bottom but becomes a highly personalized digital tool that can be worn and carried around. This marks a fundamental change in our understanding of public space and how society is organized and managed.

Empirical Analysis

Recognition: The Mediated Entry Point into Urban Identity

In today's data-driven city, whether people can be "seen" by the system has an increasingly direct impact on how they access services, exercise their rights, and socialize. Recognition is no longer just a symbolic confirmation of identity, but a daily operation deeply integrated into the technical system (Amoore, 2021; Andrejevic, 2020). In the "recognition-recording-regulation" (R-R-R) framework, recognition is the first step for artificial intelligence to intervene in urban life. At this stage, the system will use clues such as a person's appearance, behavior, or way of speaking to determine who a person is and what he is going to do (Cheney-Lippold, 2017; Gekker & Hind, 2019).

In the CCTV program "Winning with AI+", smart glasses were presented as a wearable artificial intelligence device that can recognize people, objects, and environments, pushing human-computer interaction to a more delicate level. One example comes from entrepreneur Lin Zhipeng, who introduced a pair of smart glasses designed for the hearing-impaired. This device can understand the user's expression through gesture recognition, and convert speech into text to achieve instant communication (CCTV, 2025, 13:11). This system can read people's movements and turn sound into text on the screen. This not only improves the efficiency of communication but also allows people who were originally ignored by the communication system to participate, realizing the true sense of "being seen" (Goggin & Ellis, 2015).



Figure 2. Smart Glasses Scanning Sign Language in the Presenter's Gesture Zone Using AI-Based Recognition Technology

Such a recognition process is not just a detection, but also a way of connection. As Williamson (2022) pointed out, assistive technology can help users cross places that are otherwise inaccessible, just like "spatial prostheses". In this sense, smart glasses continuously adapt to the environment through different sensory channels, translating and reconstructing the external world for users. This continuous recognition mechanism allows users to be more flexible and in control of their daily movements. We can think of it as a "software-constructed spatial experience", which coincides with the concept of "code/space" proposed by Kitchin and Dodge (2011).

The recognition function of smart glasses also extends to the wider environment. In the second half of the program, it was shown that it could recognize plants, cars, and even strangers on the street in real-time (CCTV, 2025, 42:12). This shows that AI recognition capabilities have gone beyond what the human eye can see. It is like a digital sense that constantly scans and interprets the world around it. This is consistent with Larkin (2013), who believes that infrastructure itself can function like a sense, constantly acquiring and processing external information. In this way, the urban environment is transformed into a set of data that can be understood and utilized by algorithms (Mattern, 2017; Leszczynski, 2020).

Recording: Memory Mechanisms and Data Sovereignty

In the context of AI being integrated into wearable devices, "recording" is no longer just a simple technical operation. It has become a more complex process involving how people preserve memories, store personal data, and even how they present their identity. Devices like smart glasses do more than just store what they see or hear. They also begin to help people manage their attention and memory, leaving some

cognitive tasks to technology (Mayer-Schönberger, 2009).

In CCTV's program "Winning with AI+", the recording function of smart glasses was particularly emphasized. The demonstration mentioned that the glasses have "short-term" and "long-term" memory systems (CCTV, 2025, 57:33). The short-term system is used to process current information, such as recognizing conversations or real-time scenes; the long-term system stores users' past data for recommending content or recalling historical scenes (Roller et al., 2021). In other words, this device can save video and audio information in the cloud, then extract key features and organize them using some kind of indexing method. Users can quickly find previous interaction records by simply issuing instructions (CCTV, 2025, 58:18).

Although this function sounds convenient, it also brings some deep-seated problems. Some scholars such as Zuboff (2019) and Couldry and Mejias (2019) point out that when our memories become data, these data may be analyzed, used, or even controlled by others. This is not just as simple as "transferring" memories from the brain to the machine. Once this data can be read and manipulated by algorithms, personal memories become a resource that can be used. In this case, artificial intelligence is not just a tool, but more like a digital brain that can "help you remember something" (Andrejevic, 2020).

The technical logic behind this device is also consistent with the concept of "computational memory". In this system, the device does not mechanically save all content but selectively filters, classifies, and retrieves information. This sounds like the way the human brain works, but in fact, it is subject to the logic of the enterprise and the architecture of the platform (Stiegler, 2010; van Dijck, 2013). This raises some key questions, such as: Who has the right to decide what content is saved? How to save it? Under what circumstances can it be used again? These questions are at the core of the current discussion on data sovereignty (Taylor, 2017; Milan & Treré, 2020).

Regulation: Algorithmic Governance and Urban Behavioral Orchestration

With the popularity of AI wearable devices such as smart glasses, the way algorithms affect our lives has also changed. Such devices not only enhance the user's sensory experience but also actively guide or even change the user's behavior. They are like a delicate control tool that quietly plays a role in daily life. In other words, these devices have become micro-level regulatory devices in urban operations. The so-called "algorithmic governance" refers to guiding people to make specific behaviors through data-driven programs. This governance method has been integrated into our daily decision-making process, such as recommending what routes to take, prompting what to do, and even predicting what we might do next (Yeung, 2018; Kitchin, 2021).

In CCTV's "Winning with AI+" program, the case of AI glasses demonstrated this trend. The glasses demonstrated in the program integrate multiple intelligent modules and can serve as both a tour guide and a nutrition consultant. It is connected to large language models such as "Doubao" and "Deepseek". Users only need to wear the device to access various services in one stop (CCTV, 2025, 54:43). When using it, each user is actually becoming part of this technology system. The device automatically recommends, reminds, and even guides user behavior according to the needs of different people. This is not only service optimization, but also a behavior guidance mechanism.

This mechanism also strengthens the concept of a "programmable public" (Van Dijck, Poell & De Waal, 2018). In the past, urban residents were the objects of policy management; now, many management tasks have been handed over to smart devices. Take a simple example in the program. The user asked, "What will the weather be like tomorrow?" The smart glasses immediately responded: "Hangzhou will be cloudy and turn to light rain tomorrow..."(CCTV, 2025, 56:28). This response is not just a weather report. The system collected geographic location, weather data, and personal preferences in the background, and then gave an analyzed answer. This feedback mechanism is constantly optimized through environmental perception and personal data and is also quietly shaping the behavior path.



Figure 3. Multi-Functional Smart Glasses Interface Powered by AI Models for Real-Time Urban Task Orchestration

More importantly, this device begins to "represent" the user. The program mentioned that in the future, smart glasses can develop into "digital avatars" to complete tasks, respond, and even express emotions for users (CCTV, 2025, 57:45). This means that devices will no longer be just assistants, but human-machine partners with a certain degree of autonomy. This change raises some key questions: What else can users

control? How transparent are the decisions made by the algorithm? If the device makes a mistake, who should be held responsible? These issues are becoming important discussion points for algorithmic ethics and social governance (Ananny & Crawford, 2018).

Therefore, we cannot only look at such devices from the perspective of whether they are easy to use or novel. They have actually become a new behavior management system. Devices quietly change people's behavior and thinking through environmental information, attention guidance, and data analysis. This change has also attracted researchers' attention to "algorithmic power". Many scholars point out that this type of control is difficult to detect and difficult to regulate through traditional democratic mechanisms (Gillespie, 2014; Cheney-Lippold, 2017). In other words, this type of supervision is opaque and difficult to be held accountable.

Discussion and Critical Engagement

Although the R-R-R model's three stages have different technical focuses, when combined, they reveal a deeper question: How do these devices change the structure of urban society? How do they redefine the relationship between people and the city? On the one hand, these devices do have the potential to promote inclusion. For example, in the program "Smart Terminal" (CCTV, 2025, 13:11), smart glasses can recognize sign language and convert speech into visual information. This opens up new ways of communication for the hearing-impaired and enables them to participate in urban life more conveniently. These technologies are like auxiliary organs in the digital age, filling the gap in perception and interaction between people. This also brings expectations for the future city - a space that can accept differences and support diversity with technology (Williamson, 2022).

But at the same time, this inclusion also comes with a price. In order for the device to have this recognition and translation function, the system needs to continuously collect and analyze the user's behavioral data. This means that only when individuals are "visible", that is, recognized and read by the system, can they enjoy the services brought by these technologies (Amoore, 2021). Once you enter this system, your identity, behavior, and even emotions may be continuously evaluated by the algorithm. In this way, "participation in urban life" becomes a prerequisite: you must be recognized by the system and must provide data in exchange (Cheney-Lippold, 2017). This raises concerns about "conditional citizenship". People may not only be excluded from the system but also lose their privacy due to excessive transparency. The system may record behavior, predict preferences, and even make judgments without the user's knowledge (Zuboff, 2019; Taylor, 2017).

What smart glasses reveal is not only the change in the relationship between individuals and data but also a shift in the overall logic of the urban media system. In

the past, media infrastructures - such as billboards, radio and television, or surveillance cameras - were fixed, and information was transmitted from top to bottom (McQuire, 2016). But now, devices are mobile, personalized, and can respond to users in real-time. For example, the DeepSeek function introduced in the program (CCTV, 2025, 54:43), the device is like an "external brain" that can analyze the user's environment, needs, and behavior at any time. Such devices not only convey information but also participate in and guide the rhythm of urban life (Mattern, 2017; Couldry & Hepp, 2017). The media no longer just describes the city, but "choreographs" the city.

The consequence of this change is that urban management has gradually moved away from public consultation and spatial planning and turned to the algorithmic feedback system inside the device. This is in stark contrast to Lefevre's (1991) statement that "urban space is a social process." Now, as Kitchin (2021) and Andrejevic (2020) warn, the city is increasingly like an environment "optimized" by a program. In this system, smart glasses take on the role of active calculation. They set priorities based on algorithms, and determine how people act, what to pay attention to, and where to go, all of which happens with little notice from the user.

Therefore, while AI glasses do bring conveniences, such as greater accessibility, stronger perceptual support, and better navigation experience, they are also reshaping the way we understand cities, interactions, and ourselves. The use of this technology requires more cautious design thinking and governance. We cannot just focus on whether technology "works", but also on how it shapes power structures. As Milan and Treré (2020) emphasize, truly meaningful technological inclusion must take into account both data control and system accountability mechanisms, not just the improvement of convenience.

Conclusion

This research uses the "Recognition-Recording-Regulation" (R-R-R) model as an analytical tool, starting from the application of artificial intelligence smart glasses, to explore how digital media intervenes in and reshapes the social space of contemporary cities. The study found that such devices are not only a medium for information dissemination but also a part of urban operations. They change how people are identified, recorded, and guided, thus deeply embedding digital technology into the physical structure and cultural symbols of the city.

These observations are consistent with the current research view that digital media plays an infrastructural role in urban governance (Mattern, 2017; Kitchin, 2021; Andrejevic, 2020). The R-R-R framework helps us understand the spatial and political impact of this change. It shows that smart devices and platforms are not just neutral technical tools. They subtly change the way urban space is used, people's behavior

patterns, and power relations in society. As AI systems are increasingly embedded in everyday items, such as glasses or watches, the original clear boundaries - such as the distinction between individuals and platforms, and between public and private domains - have become blurred.

Finally, this study also highlights an important need: we need a more critical perspective to study media systems in cities. This means that researchers should not only focus on the conveniences brought by technology but also on how these technologies affect social equity. We must think about data justice, infrastructure design, and civil rights together. As Couldry and Mejias (2019), Yeung (2018), and Zuboff (2019) pointed out, current technological development may be shaping a "data-based citizen" identity. The way people participate in public life is increasingly restricted by invisible algorithmic rules and platform structures. In this context, if we want to ensure fairness in urban life, we need to redesign these media systems to make them more open, transparent, and easier to monitor.

References

- Amoore, L. (2021). *Cloud ethics: Algorithms and the attributes of ourselves and others*. Duke University Press. <https://doi.org/10.2307/j.ctv1lg97wm>
- Ananny, M., & Crawford, K. (2018). Seeing without knowing: Limitations of the transparency ideal and its application to algorithmic accountability. *New Media & Society*, 20(3), 973–989. <https://doi.org/10.1177/1461444816676645>
- Andrejevic, M. (2020). *Automated media*. Routledge. <https://doi.org/10.4324/9780429242595>
- CCTV. (2025, May 16). *Winning with AI+ – Smart Terminals* [Television broadcast]. China Central Television. <https://tv.cctv.com/2025/05/17/VIDEQFSY1UbK7A4uhcfpf02y250517.shtml>
- Cheney-Lippold, J. (2017). *We are data: Algorithms and the making of our digital selves*. NYU Press. <https://doi.org/10.2307/j.ctt1gk0941>
- Couldry, N., & Hepp, A. (2017). *The mediated construction of reality*. Polity Press.
- Couldry, N., & Mejias, U. A. (2019). *The costs of connection: How data is colonizing human life and appropriating it for capitalism*. Stanford University Press. <https://doi.org/10.1515/9781503609754>
- Foucault, M. (1977). *Discipline and punish: The birth of the prison* (A. Sheridan, Trans.). Vintage Books. (Original work published 1975)
- Galič, M., Timan, T., & Koops, B.-J. (2017). *Bentham, Deleuze and beyond: An overview of surveillance theories from the panopticon to participation*. Philosophy &

Technology, 30(1), 9–37. <https://doi.org/10.1007/s13347-016-0219-1>

Gekker, A., & Hind, S. (2019). Surveillance and personalization: Understanding socio-technical assemblages of algorithmic governance. *New Media & Society*, 22(4), 639–656. <https://doi.org/10.1177/1461444819879426>

Gillespie, T. (2014). The relevance of algorithms. In T. Gillespie, P. J. Boczkowski, & K. A. Foot (Eds.), *Media technologies: Essays on communication, materiality, and society* (pp. 167–193). MIT Press.

Goggin, G., & Ellis, K. (2015). *Disability and the media*. NYU Press.

Jenkins, H. (2006). *Convergence culture: Where old and new media collide*. NYU Press.

Kitchin, R. (2021). *Data lives: How data are made and shape our world*. Policy Press. <https://doi.org/10.1332/policypress/9781529215144.001.0001>

Kitchin, R., & Dodge, M. (2011). *Code/space: Software and everyday life*. MIT Press.

Larkin, B. (2013). The politics and poetics of infrastructure. *Annual Review of Anthropology*, 42, 327–343. <https://doi.org/10.1146/annurev-anthro-092412-155522>

Leszczynski, A. (2020). Glitchy vignettes of platform urbanism. *Urban Geography*, 41(3), 419–423. <https://doi.org/10.1177/0263775819878721>

Livingstone, S. (2004). The challenge of changing audiences: Or, what is the audience researcher to do in the age of the Internet? *European Journal of Communication*, 19(1), 75–86. <https://doi.org/10.1177/0267323104040695>

Mattern, S. (2017). *Code and clay, data and dirt: Five thousand years of urban media*. University of Minnesota Press.
<https://doi.org/10.5749/minnesota/9781517902438.001.0001>

Mayer-Schönberger, V. (2009). *Delete: The virtue of forgetting in the digital age*. Princeton University Press.

McQuire, S. (2016). *Geomedial: Networked cities and the future of public space*. Polity Press.

Milan, S., & Treré, E. (2020). The rise of the data poor: The COVID-19 pandemic seen from the margins. *Social Media + Society*, 6(3).
<https://doi.org/10.1177/2056305120948233>

Plantin, J.-C., Lagoze, C., Edwards, P. N., & Sandvig, C. (2018). Infrastructure studies meet platform studies in the age of Google and Facebook. *New Media & Society*, 20(1), 293–310. <https://doi.org/10.1177/1461444816661553>

Roller, S., Dinan, E., Goyal, N., Ju, D., Williamson, M., Liu, Y., ... & Weston, J. (2021). Recipes for building an open-domain chatbot. *arXiv preprint arXiv:2004.13637*. <https://doi.org/10.48550/arXiv.2004.13637>

Stiegler, B. (2010). Taking care of youth and the generations. Stanford University Press.

Taylor, L. (2017). What is data justice? The case for connecting digital rights and freedoms globally. *Big Data & Society*, 4(2).
<https://doi.org/10.1177/2053951717736335>

Van Dijck, J. (2013). The culture of connectivity: A critical history of social media. Oxford University Press.
<https://doi.org/10.1093/acprof:oso/9780199970773.001.0001>

Van Dijck, J., Poell, T., & De Waal, M. (2018). The platform society: Public values in a connective world. Oxford University Press.
<https://doi.org/10.1093/oso/9780190889760.001.0001>

Williamson, B. (2022). Governing with platforms: Public sector reform and the insertion of data platforms into education. *Policy & Internet*, 14(1), 115–132.
<https://doi.org/10.1002/poi3.270>

Yeung, K. (2018). Algorithmic regulation: A critical interrogation. *Regulation & Governance*, 12(4), 505–523. <https://doi.org/10.1111/rego.12158>

Zuboff, S. (2019). The age of surveillance capitalism: The fight for a human future at the new frontier of power. PublicAffairs.